

Optimizing the Deployment of Tiny Transformers on Low-Power MCUs

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Abstract—Transformer networks are rapidly becoming State of the Art (SotA) in many fields, such as Natural Language Processing (NLP) and Computer Vision (CV). Similarly to Convolutional Neural Networks (CNNs), there is a strong push for deploying Transformer models at the extreme edge, ultimately fitting the tiny power budget and memory footprint of Micro-Controller Units (MCUs). However, the early approaches in this direction are mostly ad-hoc, platform, and model-specific. This work aims to enable and optimize the flexible, multi-platform deployment of encoder Tiny Transformers on commercial MCUs. We propose a complete framework to perform end-to-end deployment of Transformer models onto single and multi-core MCUs. Our framework provides an optimized library of kernels to maximize data reuse and avoid unnecessary data marshaling operations into the crucial attention block. A novel Multi-Head Self-Attention (MHSA) inference schedule, named Fused-Weight Self-Attention (FWSA), is introduced, fusing the linear projection weights offline to further reduce the number of operations and parameters. Furthermore, to mitigate the memory peak reached by the computation of the attention map, we present a Depth-First Tiling (DFT) scheme for MHSA tailored for cache-less MCU devices that allows splitting the computation of the attention map into successive steps, never materializing the whole matrix in memory. We evaluate our framework on three different MCU classes exploiting ARM and RISC-V Instruction Set Architecture (ISA), namely the STM32H7 (ARM Cortex M7), the STM32L4 (ARM Cortex M4), and GAP9 (RV32IMC-XpulpV2). We reach an average of $4.79 \times$ and $2.0 \times$ lower latency compared to SotA libraries CMSIS-NN (ARM) and PULP-NN (RISC-V), respectively. Moreover, we show that our MHSA depth-first tiling scheme reduces the memory peak by up to $6.19 \times$, while the fused-weight attention can reduce the runtime by $1.53 \times$, and number of parameters by 25%. Leveraging the optimizations proposed in this work, we run end-to-end inference of three SotA Tiny Transformers for three applications characterized by different input dimensions and network hyperparameters. We report significant improvements across the networks: for instance, when executing a transformer block for the task of radar-based hand-gesture recognition on GAP9, we achieve a latency of 0.14 ms and energy consumption of 4.92 μ J, $2.32 \times$ lower than the SotA PULP-NN library on the same platform.

Index Terms—Deep Neural Networks, Transformers, Micro-Controller Units, Edge Computing, DNN Acceleration

1 INTRODUCTION

IN the last few years, there has been a significant trend towards moving computing workload from centralized facilities towards the extreme edge of the Internet of Things (IoT), improving privacy, efficiency, and ensuring a predictable and constant latency, independent from the network coverage and congestion [1]. As a consequence, modern IoT endpoints have been changing from simple Micro-Controller Units (MCUs) equipped with the sensor infrastructure for data collection and transmission towards more complex heterogeneous Systems-on-Chip (SoCs) characterized by dedicated accelerators and enhanced Instruction Set Architectures (ISAs) to optimize efficient on-board computations – while still meeting to tight power, performance, and cost constraints. For this reason, extensive research has been conducted on how to squeeze complex Machine Learning (ML) models into such constrained devices with a strong emphasis on Deep Neural Networks (DNNs) due to their success in solving many challenging tasks. Standard hardware-agnostic optimization techniques include topological changes [2], data quantization [3], [4], and pruning [5]. These efforts are often collectively labeled as *TinyML* and are applied to State of the Art (SotA) Con-

volutional Neural Networks (CNNs), such as ResNets [6] or MobileNets [7], that dominates most of the tasks of the IoT domain, including image recognition, object detection, or time-series analysis.

At the same time, increasing efforts have been dedicated to finding a successor architecture to CNNs for IoT processing to increase tasks' accuracy further. A strong candidate is the Transformer architecture [8], which first emerged as SotA model in Natural Language Processing (NLP) and is now also SotA in other domains such as Computer Vision (CV) and Audio processing. The Transformers architecture relies on the attention mechanism that, thanks to its generality and flexibility, has enabled significant advancements in language understanding, text generation, and translation tasks [9]. Transformer models have been initially developed targeting "at scale" deployment in the cloud. Industry and academia are now increasing efforts toward Transformer models tuned for edge deployment. Recent advancements have focused on the minimization of Transformer model size [10] while still maintaining leading-edge accuracy when compared with CNN models of similar size, thus introducing a novel class of Tiny Transformers. The success of these early reduced-scale Transformer models confirms that Tiny Transformers are relevant and applicable in Edge computing scenarios.

However, the introduction of Transformers comes with novel computational primitives not commonly found in CNNs and has not been optimized so far. Precisely, the Multi-Head Self-Attention (MHSA) operation comes with many challenges, such as the high memory footprint of intermediate results and frequent data marshaling. Despite our early work demonstrating encoder-only Tiny Transformer deployment [10], the approach was highly model and platform-specific and not easily generalizable to multiple models and platforms. Moreover, early work did not

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